Listing 2.3 Use the YOLO model to demonstrate multiple object detection within a video stream.

```
import cv2
import numpy as np
from ultralytics import YOLO

def run_webcam_detection():
  model = YOLO("yolov8n.pt") #1
  cap = cv2.VideoCapture(0) #2
  cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640) #3
  cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

  while True:
    ret, frame = cap.read() #4
    results = model(frame) #5
    annotated_frame = results[0].plot() #6
    cv2.imshow("YOLOv8 Detection", annotated_frame) #7

    if cv2.waitKey(1) & 0xFF == ord('q'):
      break

  cap.release()
  cv2.destroyAllWindows()
```

#1 Use the YOLOv8 nano model, which is optimized for speed and resource efficiency.
#2 Open the webcam.
#3 Set the webcam properties.
#4 Continuously read a frame from the webcam.
#5 Run YOLOv8 inference on the frame.
#6 Visualize the results on the frame.
#7 Display the annotated frame.

Figure 2.4 demonstrates the use of the YOLO model to detect people within a room, which can be used to determine occupancy within a room by detecting people within the frame.